M KEERTHI

AI/ML Engineer | Machine Learning & RAG Applications | Python

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github.com/mkeerthi63 | Portfolio: mkeerthi63.github.io/KEERTHI/ai-ml.html

SUMMARY

AI/ML Engineer and B.Tech CSE graduate (2026) who builds machine learning and Generative AI applications from start to finish: feature engineering, model training, evaluation, tuning and deployment. Built an LLM-powered **RAG assistant** (routing accuracy 43.8% → 96.9%), a water-quality forecaster that lifted R^2 from 0.300 to 0.449 with PyTorch and MLflow, and a tuned classifier (**ROC-AUC 0.868**). Masai School AI & ML trainee and Shell-sponsored AICTE Machine Learning intern.

TECHNICAL SKILLS

Languages & Libraries: Python, SQL, PyTorch, Scikit-learn, Pandas, NumPy, imbalanced-learn, Matplotlib, Seaborn
Machine Learning: Classification, Regression, Multi-Output Regression, Random Forest, Logistic Regression, Decision Tree
Generative AI / NLP: RAG, Vector Embeddings, Semantic Search, LangGraph, ChromaDB, Sentence Transformers, Pydantic
Evaluation & Data: GridSearchCV, Cross-Validation, ROC-AUC, F1, Precision, Recall, MAE, RMSE, R^2, Imputation, ETL
MLOps & Tools: MLflow, GitHub Actions (CI), Docker, FastAPI, Streamlit, Pytest, Git, Jupyter, SQLite, MySQL, Selenium

AI/ML PROJECTS

RAG Customer Support Assistant | github.com/mkeerthi63/support_assistant *Llama 3.3, LangGraph, ChromaDB, FastAPI*

- Built a RAG assistant answering Zepto policy questions with **Llama 3.3 70B (Groq)**, grounded in 8 policy documents.
- Indexed **all-MiniLM-L6-v2** embeddings in **ChromaDB**; on 32 labeled queries, **Hit@1 91.7%, Hit@3 100%, MRR 0.958**.
- Replaced a keyword router with a hybrid semantic **LangGraph** router, lifting routing accuracy from **43.8% to 96.9%**.
- Validated LLM JSON with **Pydantic** and retries; served via **FastAPI** in **Docker**; **GitHub Actions** CI runs tests.

Survival Prediction ML Pipeline | github.com/mkeerthi63/Capstone_Project *Scikit-learn, imbalanced-learn, Pandas*

- Cleaned an 891-row, 15-feature dataset: median imputation for missing age (19.9% of rows), mode for categoricals, one-hot encoding and scaling in a **ColumnTransformer** pipeline.
- Compared Logistic Regression (ROC-AUC 0.852), Decision Tree (0.814) and Random Forest; tuned Random Forest was best with **accuracy 82.6%, F1 0.754, ROC-AUC 0.868**.
- Tuned with **GridSearchCV** (5-fold CV, F1 scoring, OOB 0.82); **SMOTE** lifted recall from 0.70 to 0.74.
- Built a fare regression model (MAE 18.87, RMSE 30.92) with residual analysis; serialized the full pipeline with **joblib**.

Water Quality Prediction | github.com/mkeerthi63/Water-Quality-Prediction *PyTorch, Scikit-learn, MLflow, Streamlit*

- Forecast 6 pollutants for 22 river stations (**2,861 samples**, 2000–2021) to flag pollution before the next lab test.
- Error analysis showed station and year could not explain O2/NO2; engineered lag, rolling-mean and seasonal features.
- Compared baseline, Random Forest and a **PyTorch MLP** on a time-based split; mean **R^2 rose 0.300 → 0.449** (CL 0.955).
- Logged every experiment in **MLflow**; built a **Streamlit** app that forecasts any station's next sample.

Book Catalog ETL Pipeline | github.com/mkeerthi63/Capstone_Project *BeautifulSoup, SQLite, SQL, Pandas*

- Built an automated ETL pipeline that scrapes book data with retry handling, cleans prices, ratings and stock status, and loads a 2-table SQLite schema.
- Wrote 5 analytical SQL queries, including a JOIN, and validated the results against Pandas *read_sql* and *merge* outputs.

EXPERIENCE

KarbonWise – *QA Automation Engineer Intern* Sep 2026 – Present | Hyderabad

- Build Python test automation with **Pytest, Selenium and Playwright** for a carbon-accounting web app.
- Validate data-ingestion workflows (Inventory Upload, Client Templates) via functional, regression and smoke tests.
- Work with developers through Git and Bitbucket pull requests and test APIs with Postman.

AICTE Virtual Internship (Edunet, sponsored by Shell) – *Machine Learning Intern* Jun 2025 | Remote

- Built the Water Quality Prediction project end to end: cleaning, feature engineering, modeling and Streamlit app.

EDUCATION

Masai School – *Artificial Intelligence & Machine Learning Program* Jan 2026 – Aug 2026
CMR Institute of Technology, Hyderabad – *B.Tech, Computer Science and Engineering* | 80% 2022 – 2026
KGBV Waddepally, Gadwal – *Intermediate (MPC)* | 87.9% 2020 – 2022
ZPHS Undavelly, Gadwal – *SSC* | 95% 2020